SoftyFy — Storage & Hosting Plan

Cloudflare R2 + Backblaze B2 + GitHub Releases (free tier, self-hosted audio)

Pure-audio music app · background playback · per-song URLs · no code change required

> **Ek line me:** teeno free hosts jodkar **~25 GB** milta hai — yaani **~3,500 HQ songs** (ya ~7,000 LQ songs) bina ek rupaya kharch. 100,000 songs free me nahi ho sakte (~1 TB chahiye).

1. Storage — Kitna aur kitne songs

Host	Free storage	HQ songs (~7 MB)	LQ songs (~3.5 MB)
Cloudflare R2	10 GB	~1,400	~2,850
Backblaze B2	10 GB	~1,400	~2,850
GitHub Releases	~5 GB	~700	~1,400
Total	**~25 GB**	**~3,500**	**~7,000**

Quality	Per song	Songs @ 25 GB
HQ (160–192k)	~7 MB	~3,500
LQ (128k)	~3.5 MB	~7,000
Very low (64k, poor)	~1.7 MB	~14,700

Free ceiling: realistically **~3,500 HQ songs** (best quality vs count balance). 64k tak gira ke ~14,700 tak push ho sakta hai, par quality kharab lagegi.

100k free me kabhi nahi — 100,000 × ~10 MB (HQ+LQ) ≈ 1 TB. Koi bhi free host itna nahi deta.

2. Kaam kaise karega (Working)

Setup (ek baar)

1. **Cloudflare R2** bucket banao — *public* access + public URL (ya custom domain) enable karo.
2. **Backblaze B2** bucket banao — *public* karo.
3. **GitHub** release tag banao (jaise `softyfy-audio-v1`) — jaise abhi chal raha hai.

Upload pipeline

4. Ek upload script se song ko target host pe bhejo — public URL milega.
5. Catalog entry me us song ka `audioSrc` = us host ka URL set karo.
6. LQ version (`audioSrcLQ`) bhi waise hi kisi host pe daalo.

> **Zero code change:** tumhari app *per-song* `audioSrc` leti hai. Isliye ek hi app me kuch songs R2 se, kuch B2 se, kuch GitHub se — bas entry ka URL alag host ka.

Distribution strategy

- **R2** → **HQ songs** (egress free — sabse zyada playback wale songs yahan).
- **B2** → **baaki / HQ overflow** (egress free 3× storage/day).
- **GitHub** → **purane chhote set / fallback** (~5 GB, throttle ho sakta hai).

Playback (runtime)

- `<audio>` element URL load karta hai; teeno hosts **206 byte-range** stream karte hain.
- Har song *load time* pe HQ ya LQ choose karta hai (manual setting > measured stalls > connection hint).
- Service worker **"play pe cache"** karta hai (offline replay) — poora library precache nahi.
- Background playback native `<audio>` se chalta hai — app ka current behaviour intact.

3. Issues / Risks (ye zaroor padho)

1. **Free limit cross toh paisa lagega.** R2 10 GB ke baad ~$0.015/GB/mo; B2 ~$6/TB/mo. Usage monitor karte raho.
2. **R2/B2 pe credit card on-file.** Free tier me charge nahi hota, par signup pe card maangte hain.
3. **Egress/bandwidth.** R2 egress free (best); B2 free sirf 3× storage/day tak, uske baad $0.01/GB; GitHub ~1 GB/hour throttle.
4. **GitHub ToS.** GitHub CDN nahi hai — heavy traffic "acceptable use" ke against. Isliye ise *fallback* rakho, primary nahi.
5. **Multi-host complexity.** 3 account, 3 script, 3 credential — maintenance zyada. Koi host URL badla toh catalog update karna padega.
6. **CORS / streaming.** `<audio>` playback ke liye CORS zaroori nahi, par R2/B2 bucket *public* + sahi Content-Type + range support hona chahiye warna error.
7. **Catalog scale (alag maamla).** Hazaron+ entries ho toh `songs.ts` ko JS bundle me nahi, off-bundle JSON + list virtualization chahiye. ~3,500 songs par ye zaroori nahi.
8. **Upload time.** ~25 GB upload me kai ghante (network speed pe depend).
9. **Legal.** YouTube se download + rehost karna YouTube ToS violate karta hai; copyrighted music distribute karna illegal. Personal use me risk kam, par flag kar raha hoon.
10. **Offline cache per device.** ~7 MB/song phone me bhar jayega — "play pe cache" hi rakho, poori library download mat karo.

> **Sabse bada trade-off:** 100k + free + pure-audio + background playback — **yeh chaar saath nahi ho sakte.** Free me realistic limit **~3,500 HQ songs** hai. Uske aage jaana hai toh ya quality ghatao (LQ/64k) ya chhota paid (~$6/mo B2) uthao.

4. Recommendation

- **Free rehna hai:** R2 + B2 + GitHub mix pe ~3,500 HQ songs — LQ/HQ, background, offline sab kaam karega.
- **Zyada chahiye:** 5,000–10,000 songs sirf B2 pe ~$0.3–0.6/mo.
- **Full 100k:** ~$6–15/mo paid storage; app architecture same, sirf URLs badalenge.
